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Evaluating and Improving Personalized Product Recommendation with Multi-Signal Machine Learning on Large-Scale Retail Data

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ABSTRACT An omnichannel retailer records purchases, web and app behaviour, prices, promotions, stock and stores, yet recommends one list to everyone: the best-sellers, or the items a member bought before. Turning these signals into a personal weekly list meets three obstacles: each signal lives in its own table with its own grain and clock, a feature may only use what was known when the recommendation was made, and the usual offline score skips the members a list misses. On eleven operational exports of a multi-country sporting-goods retailer (66.1 M sale lines, 1.0 B web events, 4.7 M buying members), a Bronze / Silver / Gold lakehouse with seven quality gates builds a point-in-time feature store with zero leakage violations. On it, a rule-based two-stage recommender scores a normalized discounted cumulative gain at 20 of 0.206 when, as usual, only the members whose candidate list holds a purchase are scored; counting every purchase, it scores 0.095, and 58 % of the missed purchases were never retrieved, almost all of them items new to the member. We therefore treat retrieval as a model: eight learned candidate sources, including linear auto-encoders, random-walk, nearest-neighbour, two-tower and self-attentive sequential models, are restricted to items new to the member, refitted every quarter in every split, tuned on the explore coverage they add, and merged by a rank-quota mixer under a cap of 300 candidates. Coverage rises from 0.370 to 0.450. Under one tuning budget, a tuned gradient-boosted ranker, a stacked variant and a deep multi-task ranker with repeat and explore heads and a learned gate are compared with five next-basket baselines. All two-stage rankers beat the strongest baseline by more than a quarter. Over the baseline ranker on the same candidates, new features give about 70 % of the gain and hyper-parameter tuning the rest; the model family adds least: the deep model gains about one percent over the tuned trees only in its single-head form, and the learned gate, not the explicit split, carries the benefit. Restricted to features a purchase log yields, the pipeline keeps about four fifths of its gain over the frequency rules a retailer uses today; the other data dimensions add the rest, mostly on new items. A controlled study confirms that naive stacking inflates in-sample metrics.

INDEX TERMS Data lakehouse, evaluation methodology, gradient boosting, learning to rank, multi-task learning, next-basket recommendation, recommender systems, retail analytics, two-stage retrieval.

I. INTRODUCTION

AN omnichannel retailer knows a great deal about its loyalty members: what they bought and returned, what they looked at on the web site and in the app, which sports they follow, and, for every product, its price, promotions, stock and season. Yet the list a member sees is usually the same for everyone, the national best-sellers, or a reminder to buy the same items again. The business goal of this work is a personal top-K list for every member, every week, and an honest number for how good it is.

Three obstacles stand between the raw data and that list. Data: each signal lives in its own operational table with

its own grain, keys and clock, and the click-stream alone holds a billion events. Time: a feature may only use what was known before the recommendation was made, otherwise offline scores promise gains that vanish in production. Measurement: a score computed only where the candidate list already holds a purchase says nothing about the members the list misses.

Problem. Given a member u at an anchor t (a Monday), predict the product models u buys in $(t, t+30]$ as an ordered top-K list; a hit is a model bought and not returned, and every feature uses only data recorded before t. A natural baseline is a rule-based two-stage recommender: five hand-written candidate rules (R1 repeat, R2 co-purchase, R3 web department, R4 sport $\times$ offer, R5 national best-sellers) propose 200 models per (member, Monday) anchor, and a LightGBM LambdaRank ranker [1], [2] (P1), tuned once over 73 Gold pair features, orders them. Scored the usual way, on the anchors whose list holds a purchase, it reaches a normalized discounted cumulative gain NDCG@20 of 0.206. Scored over every anchor with a purchase, counting purchases outside the list as misses, the same model scores 0.090 (TH + VN: 0.095), and 40.7 % of the anchors have no purchase in the list at all. That gap is the research problem.

This paper therefore builds the data foundation first and then measures what each modelling step is worth. It turns eleven operational exports into a leakage-controlled lakehouse, reformulates next-basket recommendation (NBR) as a two-stage problem in which retrieval is a model with its own objective, and compares gradient-boosted trees (GBDT), a stacked variant and a deep multi-task ranker with strong NBR baselines under one candidate list, one protocol and one tuning budget. The research questions are phrased as questions, not as expected orderings, because a GBDT win is a valid outcome [3]:

- **RQ1 (data).** Can eleven raw operational exports become a point-in-time, leakage-free feature store at member–week grain on one machine, and what does it hand to the models?
- **RQ2 (retrieval).** Scored on every purchase, where does the rule-based baseline fail, and how much do learned candidate sources raise candidate recall and explore coverage over R1–R5, at what cost, and how much of that gain reaches NDCG@20 when the ranker is unchanged?
- **RQ3 (ranking and tuning).** On the same candidates and the same budget, which ranker wins (tuned GBDT, stacking, a deep multi-task ranker with repeat and explore heads, NBR baselines), and how much of the gain comes from features, hyper-parameter tuning and the model family?
- **RQ4 (value).** How many times better than "best-sellers" and "buy the same again" is the best system, for whom, at what compute cost, what purchase value does it capture (offline proxy), and how much of its gain needs data beyond the purchase history?
- **H1 (hypothesis).** Naive stacking inflates metrics on training and pseudo-test anchors relative to a rolling protocol; on the real test the gap narrows but does not vanish.

The contributions are: (1) a Bronze / Silver / Gold lakehouse with seven quality gates and a point-in-time ledger that turns the exports into model inputs with zero leakage violations; (2) an unconditional evaluation protocol with a repeat / explore split and a positive-level error decomposition that locates 58 % of the missed purchases in retrieval rather than ranking; (3) an explore-only retrieval layer of eight learned sources refitted quarterly in every split, with a rank-quota mixer; (4) a fair-budget comparison of P1-tuned, P2, P3 and five baselines on the same candidates that separates features, tuning and model family, with a controlled measurement of stacking leakage (naive, rolling, leaky); and (5) a static demonstrator with privacy safeguards that also shows the best model's batch output on real, anonymised members.

II. RELATED WORK

Lakehouse and leakage control. The lakehouse architecture [4] keeps open columnar files as the storage layer and adds managed tables and transformations on top; DuckDB [5] runs such analytical SQL in process on one machine. Leakage, a feature that uses information unavailable at prediction time, is a common cause of inflated and irreproducible results in applied machine learning [6]; we test for it at every anchor.

Next-basket recommendation. The reality check of Li et al. [7] showed that repeat purchases dominate common NBR metrics and that simple frequency baselines (P-TopFreq, GP-TopFreq) are hard to beat; it introduced the repeat / explore recall split used here. TIFU-KNN [8] models personal item frequency with time decay and neighbour smoothing. More generally, carefully tuned simple baselines often match recent neural recommenders [9].

Item-based and graph retrieval. EASE [10] is a closed-form linear auto-encoder; RP3beta [11] re-weights random walks by item popularity; item2vec [12] learns co-occurrence embeddings.

Neural retrieval and sequential models. Two-tower models [13] score users and items in a shared space with a softmax over items; SASRec [14] applies causal self-attention [15] to the behaviour sequence, gSASRec [16] corrects its overconfidence under negative sampling, and PinnerFormer [17] trains on a dense all-action objective over a future window. DIN [18] adds target attention of the candidate over the behaviour sequence.

Multi-task and repeat-aware ranking. MMoE [19] gates shared experts per task. Repeat-aware recommenders separate repeat and explore modes (RepeatNet [20], ReCANet [21]). Our P3 gates two listwise heads with a learned member-level probability.

Leakage in stacking. Out-of-fold or temporally rolling fits are standard for stacked models [22]; we measure the inflation of an in-sample fit directly.

III. DATA FOUNDATION

This section answers RQ1: how the raw exports become model inputs, and what the processing leaves for the models. Fig. 1 gives the overview.

A. SOURCES

The data set is an export of eleven operational tables for Singapore (SG), Thailand (TH) and Vietnam (VN) from 2023-01 to 2026-10, about 230 GB raw: transactions (89.3 M sale, cancel and return lines, 29.7 M payment lines, 63.0 M transaction–member links), master data (5.7 M loyalty accounts, 4.4 M SKUs, 275 stores and web sites, a calendar), price and stock timelines (42.4 M and 36.7 M rows), 0.8 M favourite-sport events, and 999.7 M web and app events in 1 774 parts. Every table has its own grain, keys and clock. Nothing was transformed before its assumptions were measured: a full scan of every file found the facts that shaped the processing, among them 9.5 % duplicate sales rows from overlapping extracts, cancellations recorded as separate lines that net a sale to zero, sale times in local clocks while version stamps are in UTC, only 34 % of the web events carrying a member, a stock table that tracks spare parts only (the stock signal is therefore switched off), and store fixtures and carrier bags in the catalogue.

B. THROUGH THE TIERS

Bronze keeps. A data contract per table declares its key, required columns, join-key types and partition. The loader writes ZSTD Parquet partitioned by country, year and month (127 GB, full load in 1 h 51 min, the click-stream alone 87 min); rows the parser cannot read go to a quarantine table with a reason instead of being dropped (65 rows).

Silver corrects. Fifteen dbt models clean and harmonise, one technique per data problem: versioned deduplication and signed sale − cancel netting per transaction and item, with the status derived from the net amount; UTC through each unit's time zone, which removed a seven-hour shift of the online lines; EUR conversion; a member bridge (one member per transaction, staff and business accounts excluded); SKU → model mapping; national price intervals per SKU; a sales calendar inferred from peaks of the promotion share plus public holidays (SG 53, TH 106, VN 58 days); and identity stitching that links web cookies to members through web orders and logins, valid within ±90 days of the link. 5.58 M netted lines are refused with one of six reasons (mostly non-merchandise and cancelled orders) and kept in quarantine.

Gold respects time. Eight Gold models build the anchor grid (Mondays after a member's second purchase), three feature entities, member at t (recency, frequency, monetary value, price profile, web activity), model × country at t (price, promotion, season, popularity; 4.29 M rows) and pair (u, m, t) (own and department history, co-purchase, web intent), the rule candidates R1–R5, and the labels: every non-returned model bought in $(t, t + \Delta]$ with $\Delta = 30$ days ($\Delta = 90$ as an auxiliary label). Prices at an anchor come from an ASOF join on the SKU intervals. Every feature reads only data recorded before its anchor, and a web event counts only if its member link was known at the anchor.

TABLE 1. Go / no-go gates per country and year (QC-2 to QC-4) and the share of transactions linked to a member.

Country	2023	2024	2025	2026	member link
TH	PASS	PASS	PASS	PASS	86–100 %
VN	PASS	PASS	PASS	PASS	62–76 %
SG	REVIEW	REVIEW	PASS	REVIEW	50–64 %

Quality gates. Seven checks run across the tiers: null keys (QC-1), referential integrity (QC-2), value domains (QC-3), uniqueness (QC-4), volume drift (QC-5), point in time (QC-6: six singular tests and a leakage ledger in which every violating row is logged) and batch–stream reconciliation for an optional streaming lane (QC-7). The 31 dbt models carry 182 tests in five layers.

C. AFTER PROCESSING

The fact table keeps 66.1 M net sale lines of the 89.3 M raw lines, in 26.3 M transactions by 4.7 M buying members. Stitching raises the share of web events linked to a member by 6–8 points (SG 36 to 42 %, TH 34 to 40 %, VN 30 to 38 %). The 64 Gold tests pass and the QC-6 ledger holds zero point-in-time violations in all three countries. Per country and year, go / no-go gates on member link ($\geq 60\,\%$), item match ($\geq 95\,\%$), price coverage ($\geq 90\,\%$) and members with two or more transactions ($\geq 20\,000$) group the countries (Table 1): TH and VN pass every year and form the main evaluation set; SG is processed in full but flagged REVIEW, because its member link is low and its labels miss the repeat purchases of unlinked transactions. Unless stated otherwise, every number below is on the test split, TH + VN, $\Delta = 30$.

D. MODEL INPUTS AND SPLITS

The unit of prediction is a (member, anchor Monday) pair. Splits are by anchor date with a 42-day purge gap: train up to 2025-09-15, valid anchors 2025-10-27 to 2025-12-22, test anchors 2026-02-02 to 2026-07-27. Anchors are capped per country by hash rank (50 000 train, 15 000 valid, 50 000 test).

Every model-generated table carries its fit cutoff, the latest source timestamp it read and a model version, and is checked by QC-6: no row may read an event at or after its anchor, and no fit may use a cutoff later than the anchor. Learned candidate sources and signal models are refitted at 13 quarterly cutoffs (2023-07-01 to 2026-07-01) in every split; an anchor uses the latest cutoff $\leq t$. A single fit at the end of train would leave 27–34 % of the test positives outside a 5 000-model universe that is 5–10 months old; quarterly refits halve that share, to 15.5 % (TH + VN, Section V). A fit at a cutoff inside valid or test reads the purchases of that split before the cutoff, so the 42-day purge gap separates the splits for the rankers but not for the candidate sources and signals; their protection is the point-in-time rule above. Each country has its own sources, universe and history: a member's purchases in another country are not read. Rankers are trained once on

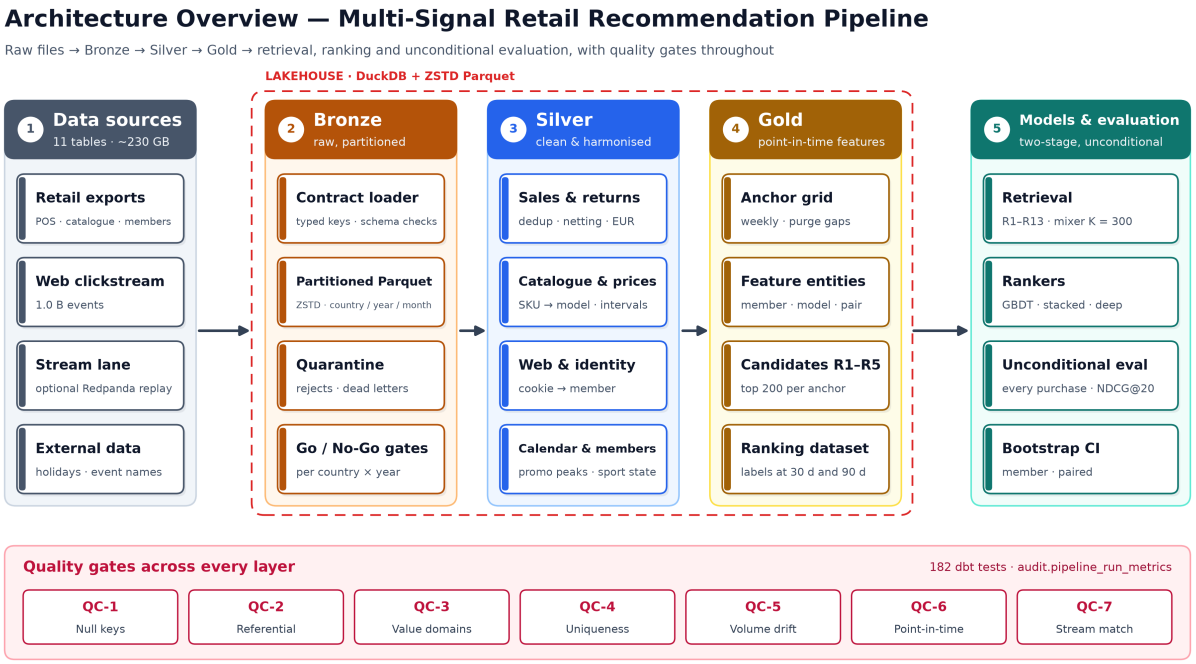


FIGURE 1. Data foundation: Bronze, Silver and Gold tiers of the DuckDB lakehouse, with the quality gates QC-1 to QC-7 across every layer.

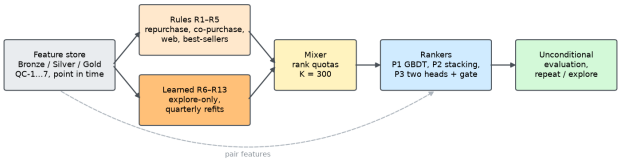


FIGURE 2. Two-stage system. The candidate sources and signal models are refitted at quarterly cutoffs before the anchors they score; the rankers are fitted once on train.

train and selected on valid. Fig. 2 shows the two-stage system that the following sections build on these inputs.

The lakehouse (DuckDB 2.0, dbt 1.12) runs on a single machine: one country on a laptop, the three countries on a 16-thread server with the same commands. The models run on one shared GH200 node (72 Arm cores and one Hopper-class GPU); deep models use bfloat16 mixed precision. Every model with hyper-parameters receives the same budget of 10 Optuna [23] TPE [24] trials on valid, which is TPE's random start-up phase, i.e., a seeded random search; the rankers were re-tuned with 30 trials as a check (Section VII) and the final fits repeated with three seeds.

IV. UNCONDITIONAL EVALUATION AND ERROR ANALYSIS

With the inputs fixed, the first half of RQ2 asks where the rule-based baseline fails when every purchase counts.

A. PROTOCOL

The evaluated unit is every used anchor a with at least one non-returned purchase in $(t, t+\Delta]$; its positives P_a are all of those purchases, including those outside the candidate list C_a, which are placed below every candidate, and those outside the per-anchor model grid that filters every candidate list (0.04 % of the positives). With L_a^K the top K of the ranked list,

$$\text{Recall@}K = \frac{|P_a \cap L_a^K|}{|P_a|}, \tag{1}$$

$$\text{NDCG@}K = \frac{\sum_{k=1}^{K} \mathbf{1}[\ell_k \in P_a] / \log_2(k+1)}{\sum_{k=1}^{\min(K,|P_a|)} 1/\log_2(k+1)}, \tag{2}$$

where ℓ_k is the item at rank k, so the ideal gain counts every positive, retrieved or not [25]. A positive is repeat if the member bought the model on or before t and explore otherwise [7] (the features read strictly before t; a purchase on the anchor Monday itself makes 0.3 % of the repeats); recall_repeat@20 and recall_explore@20 average over the anchors that have positives of that kind. We also report HR@10 (at least one hit in the top 10) and the candidate recall (share of positives anywhere in C_a). Metrics are member means; confidence intervals come from a bootstrap over members [26] (500 resamples) and model differences from a paired bootstrap over the same members (1 000 resamples).

TABLE 2. P1-frozen under the conditional and unconditional protocols (test, $\Delta = 30$).

Protocol, scope	anchors	NDCG@20	HR@10	rec. rep.	rec. exp.	cand.
conditional, TH + VN	25 538	0.206	0.368	0.826	0.218	1.000
unconditional, TH + VN	41 734	**0.095**	0.225	0.824	0.073	0.370
unconditional, all	63 574	0.090	0.217	0.797	0.065	0.351

TABLE 3. Fate of the test positives of P1-frozen over R1–R5 candidates (shares of positives).

	positives	top 10	rank 11+	capped	no source
repeat	18 251	0.654	0.332	0.013	0.002
explore	195 038	0.028	0.227	0.111	0.634
all	213 289	0.082	0.236	0.103	0.580

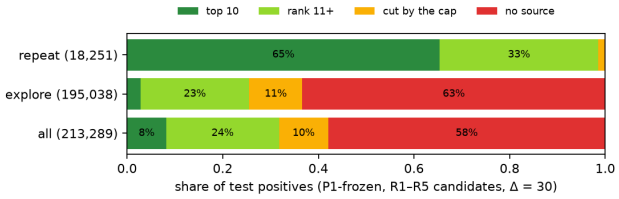


FIGURE 3. Where the test positives of the rule-based baseline end up.

B. THE RULE-BASED BASELINE UNDER THE NEW PROTOCOL

Table 2 puts both protocols side by side. The conditional protocol drops the anchors whose list holds no purchase (40.7 % of the anchors with a purchase over the three countries, 38.8 % in TH + VN), which doubles the headline number.

C. WHERE THE ERRORS ARE

Each of the 213 289 test positives is put in one cell: ranked in the top 10, ranked below 10, proposed by a source but cut by the 200 cap, or proposed by no source (Table 3, Fig. 3).

Repeat is largely solved; explore is 91 % of the positives and 74.5 % of it never reaches the ranker. Of the capped positives, 95 % come from R5 (the national top 300), whose lower ranks do not fit the cap. The positives without a source are mid-tail, not long-tail: 95 % have a popularity rank between 301 and 5 000 at the anchor. A learned source over a per-country universe of the top 5 000 models can reach most of them, but the universe is fixed at the quarterly cutoff, not at the anchor: it holds 84.5 % of the test positives (TH + VN, 83.4 % of the explore ones), which is the ceiling of every learned source (Table 6). R4 (sport × offer, recall 0.0002) is dropped: only 1.07 % of the anchors have a favourite sport. These facts set the agenda of Section V.

V. LEARNED RETRIEVAL (RQ2)

A. SOURCES

R1 (repurchase) already holds 99.7 % of the repeat positives, so every learned source is explore-only: it masks the models the member bought before the anchor and is tuned on the explore coverage it adds to R1–R5@200 on valid (Δ cand_explore of the union), not on its own recall. Each source is fitted per country at each quarterly cutoff over the top 5 000 models by buyers in the previous 365 days, and proposes 100 candidates per anchor (Table 4). With 5 000 models per country, an exact matrix product on the GPU replaces an approximate nearest-neighbour index.

TABLE 4. Learned candidate sources and their valid gain in explore coverage over R1–R5@200: best of the first 10 trials, the parameters used. EASE, TIFU-KNN, explin, RP3beta and trend ran 30 trials; the best of 30 is at most 0.001 higher. item2vec keeps its defaults (gain of the build).

Source	Model (tuned parameters)	Δ cand. exp.
R6	EASE, closed-form item–item (λ)	+0.085
R7	item2vec over baskets (defaults)	+0.074
R8	TIFU-KNN (k, r_b, r_g, α)	+0.057
R9	two-tower, softmax over explore targets (d, τ, lr, dropout)	+0.074
R10	SASRec, dense all-action loss (d, blocks, dropout, lr)	+0.085
R11	explin, windowed linear (λ, half-life)	+0.074
R12	RP3beta, random walk (α, β)	+0.088
R13	context-aware trend (window, weights)	+0.069

The two neural sources are adapted to the target. The two-tower model (R9) draws several reference days per member and epoch, adds the month of t to the member tower, and uses a softmax over the universe that masks the models already bought. SASRec (R10) takes one step per basket (the mean of its model embeddings) with a gap embedding, and is trained with a dense all-action explore loss in the spirit of PinnerFormer [17]: each step predicts every explore model of the following Δ days. R11 (explin) is a supervised windowed linear model $\hat{Y} = XB$ with closed-form ridge solution, where X is the time-decayed purchase history before t and Y the explore purchases in $(t, t + \Delta]$. R13 reweights the recent country popularity by the member's departments and price band.

B. MIXER AND THE CAP

The ranker receives at most K candidates. The mixer interleaves the sources by weighted rank quotas: candidate i gets the key

$$\kappa(i) = \min_{s\,:\,i \in C_s} \frac{r_s(i) - 0.5}{w_s}, \tag{3}$$

where $r_s(i)$ is its rank in source s and w_s the source's standalone valid Recall@100, and the K smallest keys are kept. The weight is a hits-per-slot heuristic (all positives, three countries), not a parameter tuned for the ranking objective: it gives R1 0.106, below every learned source (0.13–0.20) and R5 (0.21), although repeats carry most of the NDCG (Table 7); Section VII tests a repeat-first variant. A per-member variant scales w_{R1} by the member's repeat share; it ties with the fixed policy on valid (TH + VN coverage at 300: 0.4841 against 0.4847), so the fixed policy is used. $K = 300$ is chosen from the recall–cost curve (Fig. 4).

C. COVERAGE

Coverage rises from 0.370 to 0.450 at $K = 300$ (Table 5), meeting the plan's target of 0.45; explore coverage rises 1.30

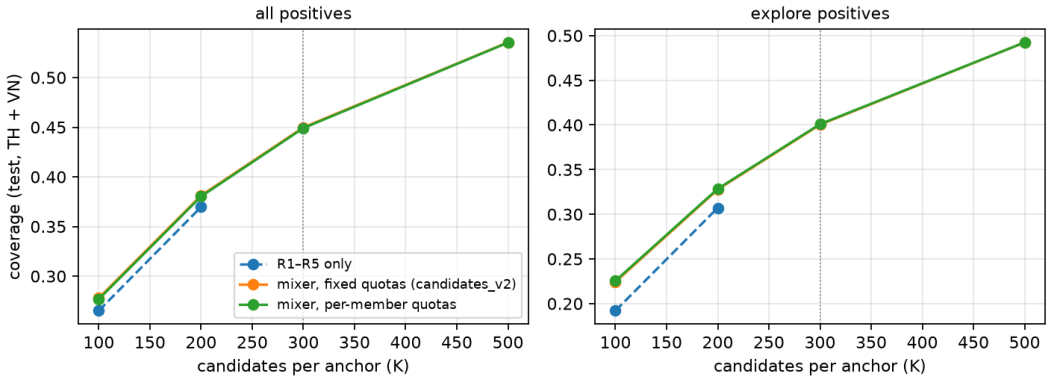


FIGURE 4. Candidate coverage against the cap K; the fixed and per-member mixers overlap at every K.

TABLE 5. Candidate coverage on test (TH + VN, $\Delta = 30$).

Candidate set	list	cand.	repeat	explore
R1–R5 (reference)	200	0.370	0.997	0.307
R1–R5 ∪ R12 RP3beta (best single)	277	0.439	0.997	0.386
R1–R5 ∪ R6 / R10 / R7	260–282	0.430–0.435	0.997	0.375–0.381
R1–R5 ∪ all learned (uncapped)	483	0.541	0.997	0.499
candidates_v2, mixer, $K = 300$	300	**0.450**	0.972	**0.400**
mixer, $K = 500$	470	0.535	0.994	0.492

TABLE 6. Fate of the test positives with candidates_v2 ($K = 300$) and P3 one head (TH + VN, shares of positives). "No source" is split by whether the model is inside the 5 000-model universe of the anchor's cutoff.

	positives	top 20	rank 21+	capped	no source in univ.	no source outside
repeat	11 533	0.785	0.151	0.043	0.019	0.002
explore	126 810	0.073	0.293	0.099	0.383	0.153
all	138 343	0.132	0.281	0.094	0.352	0.140

times (0.307 to 0.400), short of the target of 2. The cap costs 2.5 points of repeat coverage: R1 is short for most anchors (median 12 models, 90th percentile 41), but the tail of long R1 lists loses its slots to explore candidates. The sources are complementary: the union of all eight adds 0.17 over R1–R5 while the best single source adds 0.07. The neural sources do not beat the cheap ones: SASRec ties with EASE, the two-tower model is below, and RP3beta, the best single source, trains in minutes against about six GPU-hours of tuning and building for each neural source (Section VIII). This ranking is partly a budget effect: the neural sources train for a fixed 8 epochs without early stopping, and their training loss still falls by about 0.2 % per epoch at the end. Refitting them on TH at the valid cutoff with 4, 8 and 16 epochs gives Δ cand. exp. 0.074 / 0.077 / 0.082 for the two-tower model and 0.077 / 0.084 / 0.090 for SASRec: both are under-trained at 8 epochs, and twice the epochs add 0.006 of explore coverage (+8 %), more than the gap between SASRec and RP3beta in Table 4. The candidates and rankers of this paper keep the 8-epoch sources. QC-6 on candidates_v2 finds no row that reads an event at or after its anchor and no fit cutoff after the anchor. The pairs of candidates_v2 are rendered from the same dbt SQL as Gold and match it on a parity day up to floating-point noise.

What is left. Table 6 redoes the error analysis of Table 3 on the new lists. The cap is no longer the main loss: 9.4 % of the positives are proposed and cut, and R5 proposes 42 % of those (it was 95 % with R1–R5). The ceiling is the universe: it holds 84.5 % of the positives (85.9 % as a member mean), and 86.0 % are reachable at all (inside it or proposed by a rule source), against a candidate recall of 0.450. Of the remaining gap, 35 % of the positives are inside the universe but proposed by no source, a retrieval problem, and 14 % are outside it, which no learned source over this universe can fix. The 4.3 % of repeats that are capped come mostly from long R1 lists cut by the quotas (no learned source proposes them: they are masked as explore-only).

D. FROM RECALL TO NDCG

With the ranker frozen (P1-frozen's hyper-parameters and original feature set, refitted on the v2 pairs), the retrieval gain reaches NDCG@20 only partly. The paired difference v2 − v1 is $+0.0025$ $[+0.0018, +0.0032]$ in NDCG@20 and $+0.0052$ in HR@10, with $+0.0101$ recall_explore@20 and -0.0315 recall_repeat@20. The gain is significant but small (0.0947 to 0.0973): the frozen ranker has no feature that tells it which learned source proposed a candidate, it trades repeat recall for explore recall, and most new explore candidates land below rank 20.

VI. RANKING MODELS

All rankers score the same candidates_v2 lists ($K = 300$) and are selected on unconditional valid NDCG@20 with 10 TPE trials; the selection objective averages the three countries (SG included), the tables report TH + VN. Training is listwise and so uses only the train anchors with a positive in their list: 44 795 of the 150 000 sampled anchors (30 %, 13.2 M pairs). The rankers are thus trained on the conditional population and selected and evaluated on the unconditional

one.

- **P1-frozen**: the baseline LambdaRank of Section IV, with its hyper-parameters (tuned once on R1–R5 candidates) and its 73 Gold pair features kept fixed, refitted on the v2 pairs; a reference row, not a tuned competitor.
- **P1-tuned**: LightGBM LambdaRank over every pair column of candidates_v2 (85 features: the 73 of the baseline and 12 new columns, among them the ranks of the eight learned sources R6–R13).
- **P2**: P1-tuned plus five stacked signal columns from three models fitted with the rolling protocol: a LightGBM department hazard over member × department rows, $P(\text{buy in } (t, t+\Delta])$, and two LightGBM quantile models of willingness to pay (log unit price, quantiles 0.5 and 0.9), with the gaps between the candidate's log price and each quantile.
- **P3 (UMTR)**: a Transformer over the last 50 purchase lines (one token per member × model × day, not grouped into baskets; model and log-gap embeddings), an MLP over the member context, an item tower over the candidate's pair features, and DIN-style target attention [18] of each candidate over the sequence states; 3.7 M parameters.
- **Baselines**: P-TopFreq, GP-TopFreq [7], EASE and TIFU-KNN as full-catalogue NBR rankers with their own 10-trial budget on valid NDCG@20, and SASRec with the R10 network and parameters in two forms: the R10 model with the explore mask removed at scoring (it was trained on explore targets only), and the same network retrained on every model of the window, repeats included.

P3 has two heads, repeat and explore, each with a softmax over its group of candidates (R_a: models bought before t; E_a: the others). A gate $g(u) \in (0,1)$ computed from the member representation splits the probability mass:

$$p(i) = \begin{cases} g(u)\,\dfrac{e^{s_i^r}}{\sum_{j\in R_a} e^{s_j^r}}, & i \in R_a, \\ \big(1-g(u)\big)\,\dfrac{e^{s_i^e}}{\sum_{j\in E_a} e^{s_j^e}}, & i \in E_a. \end{cases} \tag{4}$$

When one group is empty (a member with no history has $R_a = \emptyset$; a list may have $E_a = \emptyset$), the gate is set to 0 or 1 and the other group takes the whole mass. The loss is listwise, $\mathcal{L} = -\frac{1}{|P_a \cap C_a|} \sum_{i \in P_a \cap C_a} \log p(i) + \lambda_g \,\mathrm{BCE}(g, y^{rep}) + \lambda_{90}\,\mathcal{L}_{90}$, where y^{rep} is the repeat share of the member's positives and $\mathcal{L}_{90}$ an auxiliary binary cross-entropy on the $\Delta = 90$ label. Early stopping uses unconditional valid NDCG@20. The one-head ablation replaces the two group softmaxes by one softmax over C_a.

VII. RESULTS

A. RQ3: MODEL COMPARISON

Tables 7 and 8 and Fig. 5 are read as answers rather than as a leaderboard.

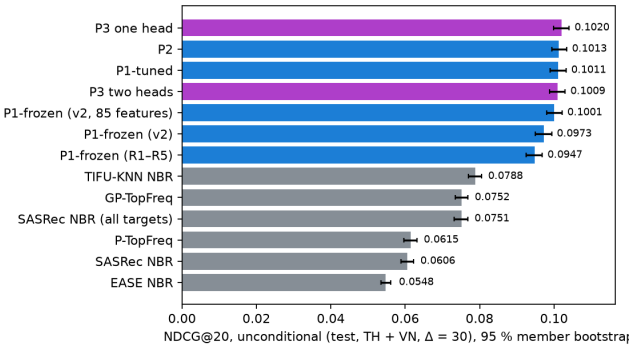


FIGURE 5. NDCG@20 of every model on test with 95 % member-bootstrap intervals.

- Every two-stage ranker beats every NBR baseline by a wide margin (P1-tuned − TIFU-KNN: $+0.022$, 28 %). The strongest baseline is TIFU-KNN, then GP-TopFreq; EASE and SASRec as full-catalogue rankers give up the repeats (recall_repeat 0.37–0.57 against 0.88). For the R10 SASRec this is partly by construction, since it was trained on explore targets only and its explore mask is only removed at scoring: retrained on every model of the window, SASRec gains $+0.0146$ $[+0.0135, +0.0157]$ NDCG@20 (repeat recall 0.43 to 0.57) and ties with GP-TopFreq, still -0.0036 $[-0.0049, -0.0025]$ below TIFU-KNN. The conclusions on the baselines do not change.
- Features matter more than tuning or the model family. P1-tuned gains $+0.0039$ over P1-frozen on the same pairs, but most of it comes from the new features, not from tuning: the frozen hyper-parameters with the 85 features already give $+0.0028$ (mostly explore recall, $+0.007$), and tuning adds $+0.0011$ (mostly repeat recall, $+0.024$). The tuning share is the size of the gap between P3 one head and P1-tuned ($+0.0009$).
- The deep model and the GBDT are within a thousandth. P3 with one head is the only model significantly above P1-tuned ($+0.0009$, about 1 %), and only just above P2. The two-head P3 ties with P1-tuned: it keeps more repeats ($+0.029$ recall_repeat@20) and finds fewer explore purchases (-0.003).
- Stacking adds nothing measurable (P2 − P1-tuned $+0.0002$, interval across zero), consistent with Section VII-D: the Gold pair features already hold department recency, cycles and price gaps.

On valid, over the three countries of the tuning objective, the order is the same (P3 one head 0.1049, P3 two heads 0.1037, P2 0.1034, P1-tuned 0.1033).

Seed variance. The final fits of P1-tuned and both P3 variants were repeated with seeds 1 and 2 at the same hyper-parameters (Table 9, Fig. 6). The seed spread (≤ 0.0003) is below the gap between P3 one head and P1-tuned; the seed-matched paired differences are $+0.0009$, $+0.0010$ and $+0.0006$, and one head beats two heads at every seed

TABLE 7. Ranking on test (TH + VN, 41 734 anchors, $\Delta = 30$, unconditional), with 95 % member-bootstrap intervals of NDCG@20. Selection on valid with 10 trials; final fits with seed 0 (seeds 1 and 2 in Table 9). P1-frozen (v2, 85 features) keeps the frozen hyper-parameters and round count but reads every pair column of P1-tuned.

Model	NDCG@20	95 % CI	HR@10	recall_repeat@20	recall_explore@20	cand. recall
P-TopFreq	0.0615	[0.0597, 0.0632]	0.156	0.877	0.000	0.093
GP-TopFreq	0.0752	[0.0734, 0.0769]	0.184	0.878	0.041	0.262
EASE NBR	0.0548	[0.0535, 0.0561]	0.146	0.368	0.072	0.231
TIFU-KNN NBR	0.0788	[0.0770, 0.0806]	0.187	0.885	0.043	0.234
SASRec NBR (R10, explore-trained)	0.0606	[0.0589, 0.0622]	0.156	0.435	0.067	0.235
SASRec NBR (all targets)	0.0751	[0.0732, 0.0769]	0.180	0.571	0.068	0.245
P1-frozen (R1–R5 candidates)	0.0947	[0.0925, 0.0968]	0.225	0.824	0.073	0.370
P1-frozen (v2)	0.0973	[0.0949, 0.0994]	0.230	0.793	0.083	0.450
P1-frozen (v2, 85 features)	0.1001	[0.0981, 0.1022]	0.236	0.793	0.090	0.450
P1-tuned (v2)	0.1011	[0.0990, 0.1033]	0.239	0.817	0.092	0.450
P2 (v2)	0.1013	[0.0994, 0.1034]	0.240	0.806	0.093	0.450
P3 two heads	0.1009	[0.0988, 0.1029]	0.237	0.845	0.089	0.450
P3 one head	**0.1020**	[0.0999, 0.1040]	0.240	0.838	0.090	0.450

TABLE 8. Paired member-bootstrap differences on test (TH + VN), $a - b$ [95 % CI].

$a - b$	NDCG@20	rec. explore@20
P1-tuned − P1-frozen (v2)	+0.0039 [+0.0032, +0.0045]	+0.0085
new features: 85 feat. − P1-frozen (v2)	+0.0028 [+0.0021, +0.0035]	+0.0069
tuning: P1-tuned − 85 feat.	+0.0011 [+0.0005, +0.0016]	+0.0016
P2 − P1-tuned	+0.0002 [−0.0002, +0.0006]	+0.0007
P3 two heads − P1-tuned	−0.0002 [−0.0009, +0.0005]	−0.0031
P3 one head − P1-tuned	+0.0009 [+0.0002, +0.0016]	−0.0016
P3 one head − P2	+0.0007 [+0.0000, +0.0014]	−0.0023
P1-tuned − TIFU-KNN NBR	+0.0224 [+0.0214, +0.0234]	+0.0488

TABLE 9. Seed variance of the final fits (test, TH + VN, NDCG@20).

Model	seeds 0 / 1 / 2	mean ± sd	rec. repeat mean ± sd
P1-tuned	0.1011 / 0.1013 / 0.1016	0.1013 ± 0.0002	0.809 ± 0.007
P3 two heads	0.1009 / 0.1005 / 0.1011	0.1008 ± 0.0003	0.824 ± 0.030
P3 one head	0.1020 / 0.1022 / 0.1022	0.1022 ± 0.0001	0.831 ± 0.008

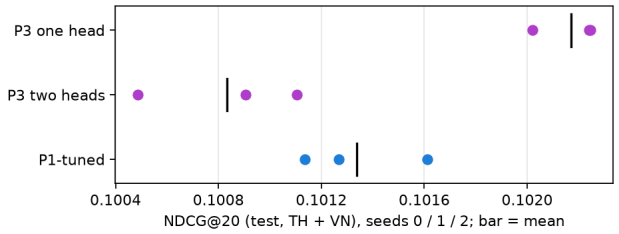


FIGURE 6. NDCG@20 of the final fits across three seeds.

(+0.0011, +0.0018, +0.0011). The two-head model is the least stable: its repeat recall moves from 0.79 to 0.85 across seeds, i.e., the learned gate lands in different places from one run to the next.

Is the cap binding? Cutting the mixer list at $K = 500$ instead of 300 raises test coverage from 0.450 to 0.535 for 1.6 times the pairs. P1-frozen was refitted as is, P1-tuned and P3 were re-tuned with 10 trials. On valid (TH + VN), the decision data, NDCG@20 does not improve: P1-tuned 0.1088 to 0.1079, P3 two heads 0.1087 to 0.1092, P3 one head 0.1096 to 0.1097, so $K = 300$ is kept. Reported once afterwards on test (paired, $K = 500$ − $K = 300$): P1-frozen -0.0009 $[-0.0016, -0.0003]$, P1-tuned -0.0003 $[-0.0008, +0.0001]$, P3 two heads $+0.0006$ $[-0.0000, +0.0011]$, P3 one head $+0.0012$ $[+0.0007, +0.0017]$ (0.1032). The one-head test gain is not supported on valid and is not used for selection, which would select on test. The GBDT loses 0.025 repeat recall to the extra negatives. The cap is not what limits accuracy; ranking the hard explore candidates is.

Are the mixer weights binding? The weights are not tuned for NDCG and cost 2.5 points of repeat coverage (Section V). A variant that places the whole R1 list first and fills the rest with the same quotas restores the repeat coverage (0.972 to 0.997 on test) for -0.001 explore coverage. P1-tuned refitted on its pairs with the same hyper-parameters gives the same valid NDCG@20 (TH + VN: 0.1087 against 0.1088), so the fixed weights are kept; on test the paired difference is $+0.0004$ $[-0.0001, +0.0009]$. The repeats the quotas cut sit at the end of long R1 lists and rarely reach the top 20. A learned mixer (LightGBM over the source ranks, top 300 of the union) lifts valid coverage to 0.502, but P1-tuned on its pairs gives 0.1084 on valid and $+0.0004$ $[-0.0002, +0.0009]$ on test.

Is the tuning budget binding? With 10 trials the best GBDT trial was the last one and sat near the learning-rate floor, and the best P3 trial sat at its floor. P1-tuned, P2 and P3 were therefore re-tuned on the same v2 pairs with 30 trials each and lower floors (GBDT learning rate ≥ 0.005 with up to 6 000 rounds, P3 learning rate $\geq 2 \cdot 10^{-5}$), the same budget for every ranker. On valid (TH + VN, the decision data), NDCG@20 does not improve from 10 to 30 trials: P1-tuned 0.1088 to 0.1089, P2 0.1089 to 0.1088, P3 two heads 0.1087 to 0.1086, P3 one head 0.1096 to 0.1092, so the 10-trial fits are kept. Reported once afterwards on test (paired, 30 − 10 trials): P1-tuned $+0.0001$ $[-0.0003, +0.0006]$, P2 $+0.0007$ $[+0.0003, +0.0011]$ (0.1020), P3 two heads -0.0002 $[-0.0005, +0.0001]$, P3 one head -0.0000 $[-0.0003, +0.0003]$ (0.1020). The search had converged at 10 trials, and the order of Table 7 stands: with 30 trials P2

TABLE 10. NDCG@20 on test by country and segment.

Model	SG	TH	VN	new	mid	loyal
TIFU-KNN NBR	0.069	0.085	0.072	0.065	0.080	0.078
P1-frozen (v1)	0.082	0.101	0.088	0.079	0.094	0.095
P1-tuned (v2)	0.089	0.108	0.093	0.089	0.101	0.099
P3 two heads	0.089	0.107	0.093	0.088	0.101	0.100
P3 one head	0.090	0.109	0.094	0.090	0.101	0.100

TABLE 11. P3 ablation, paired differences against P3 with two heads and a learned gate (test, TH + VN).

Variant − two heads	NDCG@20	HR@10	rec. rep.	rec. exp.
one head	+0.0011	+0.0025	−0.0073	+0.0016
fixed gate	−0.0066	−0.0105	−0.2753	+0.0159
no sequence	−0.0003	+0.0001	−0.0240	+0.0020

NDCG@20 intervals: one head [+0.0007, +0.0016]; fixed gate [−0.0074, −0.0058]; no sequence [−0.0007, +0.0000]. Fixed gate: g = the member's historical repeat share.

ties with P3 one head on test, but it does not on valid.

B. BY COUNTRY AND SEGMENT

Table 10 splits NDCG@20 by country and by member segment (transactions in the 365 days before the anchor: new ≤ 1, mid 2, loyal ≥ 3; all countries). The ordering holds in every country and segment, and the differences between P1-tuned and P3 stay below 0.002 everywhere. The largest gain of the whole pipeline over the reference is on new members (0.079 to 0.090, +14 %), whose purchases are almost all explore and who benefit most from the learned sources. SG (REVIEW scope) gains as much as TH and VN. A web / no-web split is not reported: 2–19 % of the anchors have web events and the web features carry about 1 % of the SHAP mass (Section VII-E).

C. RQ3: REPEAT AND EXPLORE HEADS

The explicit split does not improve explore (Table 11, Fig. 7): the single head is better on NDCG@20 and on explore recall, at a small cost in repeat recall. The gate is the part that matters: replacing the learned g by the member's historical repeat share loses 0.0066 NDCG@20 and a third of the repeat recall. A plausible reading is that the historical share (about 10 % of positives) under-weights the repeat group for the anchors where a repeat is due, which the learned gate can see from the context. The learned gate ranks segments correctly (mean g: loyal 0.23, mid 0.17, new 0.14) but is not calibrated: it is about twice the observed repeat share (0.12, 0.08, 0.06) and correlates only weakly with the per-anchor target (0.11–0.19). The sequence encoder adds repeat recall ($+0.024$) but no NDCG: the pair features already summarize the history.

D. H1: STACKING LEAKAGE

The department hazard is fitted three ways: rolling (quarterly cutoffs, the only point-in-time-correct variant), naive (fitted once on train and applied in-sample) and leaky (fitted on every split). Every fit samples at most 120 000 anchors, so

TABLE 12. Stacking leakage: unconditional NDCG@20 of a fused ranker by signal protocol (TH + VN).

	v1 candidates			v2 candidates		
Signals	train	pseudo	test	train	pseudo	test
none	0.1219	0.1072	0.0941	0.1246	0.1097	0.0961
rolling	0.1227	0.1090	0.0945	0.1255	0.1097	0.0964
naive	0.1246	0.1095	0.0934	0.1281	0.1126	0.0959
leaky	0.1232	0.1087	0.0944	0.1256	0.1091	0.0975

that the variants differ in which anchors they see, not in how many: naive samples 80 % of the 150 000 train anchors, leaky 120 000 of the 345 000 anchors of the three splits, i.e., about 35 % of the train anchors. The pseudo-test is the last train quarter (anchors from 2025-07-01 to the end of train), inside the naive and leaky fits but after the training anchors of the fused ranker below. Its AUC on train / pseudo-test / test is 0.681 / 0.703 / 0.708 for rolling, 0.725 / 0.744 / 0.705 for naive, and 0.710 / 0.717 / 0.723 for leaky. The in-sample fit looks 0.04 AUC better on the anchors it has seen and loses that lead on test; the leaky fit stays better on test because it has seen those labels.

Fused into a ranker (P1-frozen parameters, fitted on train anchors up to 2025-05-31), the signals give the NDCG@20 of Table 12. H1 holds in direction on both candidate sets, more clearly on v2: the naive fusion is ahead of rolling on its own training anchors ($+0.0026$) and on the pseudo-test ($+0.0029$), and behind it on test (-0.0004). The leaky fusion is the only one ahead on test ($+0.0011$ on v2), because its signals were fitted on the test labels. On train and pseudo-test the leaky fusion is below the naive one although it also fits in-sample: it has seen the labels of about 35 % of those anchors against 80 % for naive, so its in-sample inflation is smaller. The inflation is small because the signals are weak given the Gold features, and the gaps are of the order of the member-bootstrap intervals (± 0.002 on test, ± 0.005 on the 5 337 pseudo-test anchors). An in-sample stacked signal would nevertheless have been reported as a gain that does not exist on test.

E. WHAT THE GBDT USES

Grouped TreeSHAP [27] over 3 000 test anchors (Table 13) shows that the learned source ranks are the second feature group of P1-tuned (17 % of the mean absolute SHAP value), which is why the tuned rankers convert more of the retrieval gain than the frozen one, which does not see them. In P2 the five stacked signals take 2.4 % of the mass.

VIII. VALUE, COST AND DEMONSTRATOR (RQ4)

A. VALUE OVER SIMPLE RULES

The best model on valid, P3 with one head, is compared with three references on test HR@10 (Table 14). "Random" draws 10 slots uniformly from the 5 000-model universe (analytic); "popular" is the country top 10 by buyers in the 28 days before the anchor, the same list for every member; "buy-again" is P-TopFreq. The model is 1.8 times the best-sellers

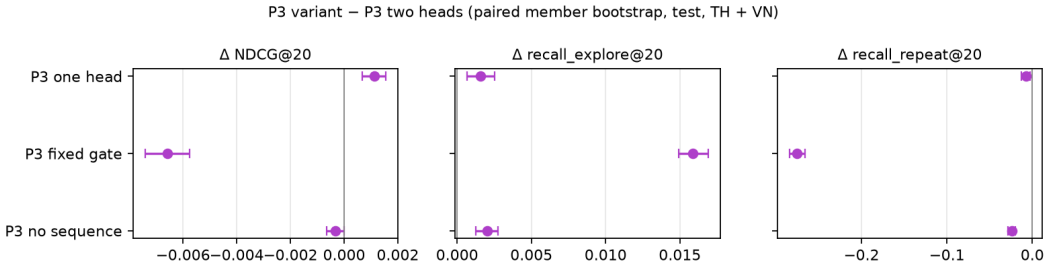


FIGURE 7. P3 ablation on test.

TABLE 13. P1-tuned: share of mean $|\mathrm{SHAP}|$ by feature group (test).

Feature group	all	repeat cand.	explore cand.
model: popularity / season / stock	0.475	0.298	0.492
retrieval: learned source ranks	0.165	0.202	0.162
retrieval: R1–R5 flags / scores / rank	0.085	0.140	0.079
member: price profile	0.080	0.031	0.085
pair: own history	0.027	0.155	0.014
web behaviour	0.013	0.013	0.013

TABLE 14. HR@10 of P3 one head against business references (test, TH + VN).

Positives	P3	random	popular	buy-again	× pop.	× again
all	0.240	0.007	0.137	0.156	1.76	1.54
repeat	0.775	0.003	0.178	0.777	4.36	1.00
explore	0.110	0.006	0.115	0.000	0.96	—

and 1.5 times "buy the same again" overall. On repeats it matches buy-again (both about 0.78); on explore purchases it is slightly below the best-sellers (0.110 against 0.115). Its value is that it does both at once in one list of ten, which no single rule does.

As an offline proxy, not a causal estimate, the GMV-weighted Recall@10 is 0.077 against 0.053 for buy-again and 0.023 for popular, i.e., 268 EUR of observed next-30-day purchase value per 1 000 recommendations against 184 and 79. The denominator counts only the anchors with a purchase in the window (41.7 % of the test anchors); per 1 000 recommendations to every anchor, buyers or not, the figures are 112, 77 and 33 EUR. These purchases happened anyway; an online test is needed to know how many a recommendation would cause.

Purchase history only. The status quo recommends from purchase history alone. To separate what the extra data dimensions bring from what the two-stage pipeline brings, P1 was re-tuned (10 trials) and refitted on the same pairs with only the 39 of 85 features a purchase log yields: R1, R2, R5 and the learned source ranks, own and department history, co-purchase, sales popularity, season and model age, activity counts and country. Web behaviour (R3 and the mixed candidate rank), prices and promotions, stock, stores, calendar, sport affinity and the member price profile are removed. Table 15 crosses data and method. The pipeline on history alone gains $+0.0214$ $[+0.0204, +0.0225]$ NDCG@20 over GP-TopFreq; the other dimensions add $+0.0045$ $[+0.0039, +0.0052]$ for P1-tuned and $+0.0054$ $[+0.0046, +0.0062]$ for P3 one head over P1-hist (paired, test; valid gives the same order: 0.0819, 0.1044, 0.1088, 0.1096). Of the $+0.0268$ of P3 one head over GP-TopFreq, about four fifths are therefore reached by retrieval and a learned ranker on the purchase log, and one fifth needs the other data, mostly as explore recall (P1-tuned − P1-hist $+0.0077$, repeat -0.0086). The history-only row is slightly optimistic: its candidate pool is the multi-dimensional one (it still holds the web-intent candidates, without the features that explain them), and the two-tower and trend sources use the unit price paid on the sales lines.

TABLE 15. Data × method on test (TH + VN, NDCG@20). P1-hist: P1 re-tuned on the 39 purchase-history features.

Data	Method	Model	NDCG@20
purchase history	frequency rule	GP-TopFreq	0.0752
	NBR	TIFU-KNN	0.0788
	pipeline	P1-hist	0.0966
multi-dimensional	pipeline	P1-tuned	0.1011
		P3 one head	**0.1020**

B. SERVING COVERAGE

Every evaluated test anchor gets a personalized list of 300 candidates (median 300; all anchors have at least 10). This holds by construction, because the anchor grid starts after a member's second purchase, so it does not answer RQ4. Over the real sales of the test window, identified members make 93 % of the GMV in TH and 81 % in VN, and members with a purchase before the sale day, who get the personalized path, make 71 % and 56 % (68 % and 58 % of the buying members). The rest, anonymous sales and first purchases, gets the country best-sellers: the models of this paper serve about two thirds of the sales value.

C. COST

Table 16 lists the cost of the candidate sources on the shared GH200 (wall times are upper bounds). The neural sources cost 50–100 times more compute than EASE or RP3beta for

TABLE 16. Cost of the candidate sources (13 cutoffs × 3 countries).

Source	device	tune h	build h	Δ exp.	per build h
R6 EASE	GPU	0.04	0.07	+0.085	1.27
R12 RP3beta	GPU	0.05	0.08	+0.088	1.16
R8 TIFU-KNN	GPU	0.04	0.08	+0.057	0.75
R13 trend	CPU	0.03	0.13	+0.069	0.54
R11 explin	GPU	0.07	0.47	+0.074	0.16
R7 item2vec	CPU	—	0.56	—	—
R9 two-tower	GPU	2.48	3.05	+0.074	0.02
R10 SASRec	GPU	4.19	3.18	+0.085	0.03

the same or a lower coverage gain. Among the rankers the costs are all small at this scale (13.2 M training pairs from the 44 795 train anchors with a positive): tuning takes 0.22–0.38 h, a final fit 1.3–5.1 min, and P3 (3.7 M parameters) scores 1 000 test anchors in 0.09 s on the GPU.

D. DEMONSTRATOR

A static page on a public Hugging Face Space runs the recommender in the browser (Pyodide). The user uploads a purchase file (member, country, model, purchase date, quantity); the page validates it and returns the top K of each member with the source of each item (repeat with time decay, co-purchase from EASE neighbours, recent country trend), merged with the same rank-quota rule as the mixer, and a reason in words. Model identifiers are salted hashes, departments carry generic labels, popularity is shown as a percentile, the sample members are synthetic, and the salt never leaves the server. This part uses reference tables frozen at 2026-07-28 and shows the retrieval layer and its explanations.

The best model needs inputs an uploaded purchase file does not hold: the web, price, promotion, stock and store features of the Gold layer and the ranks of eight learned sources, two of them (and P3 itself) PyTorch networks that do not run in the browser. It is therefore served as a real retailer would serve it, by batch scoring in the warehouse. A second tab shows that batch output: for 500 test anchors drawn at random (250 per country, one per member), the top 10 of P3 one head next to the top 10 of GP-TopFreq, the member's purchases before the anchor, and the models she bought in the next 30 days. On this sample the share of members with at least one of the ten bought is 0.240 for P3 against 0.190 for GP-TopFreq (whole test: 0.240 against 0.184). The sample is published with the data owner's consent and minimised: members get sequential identifiers in shuffled order, models the salted hashes and departments the generic labels of the reference, dates become days before the anchor, the history is cut to the last 40 lines, and no price, store, web or member attribute is included. Every deployment of the Space first scans the published files, the batch output included, for raw model codes and member identifiers, and locally for a list of organisation names.

IX. DISCUSSION AND LIMITATIONS

GBDT against deep models. Under the same budget and the same candidates, the deep multi-task ranker gains about 1 % NDCG@20 over a tuned GBDT and only in its single-head form, in line with the tabular-data literature [3]. The effort paid most in retrieval (+22 % relative coverage) and in features that let the ranker read it (+3 % NDCG@20 for the learned source ranks and the other new pair columns under the frozen parameters); tuning added 1 %.

Method against data. Most of the gain over the status quo comes from the method, not from the extra data: on purchase history alone the pipeline reaches 0.0966 against 0.0752 for GP-TopFreq, and the web, price, promotion, stock and store features add 0.005 NDCG@20 on top. A retailer with only a sales log can therefore take most of the benefit; the other dimensions matter for the explore purchases.

Explore is still the bottleneck. Explore coverage rose 1.30 times, not 2, and at the top 10 the model is no better than the best-sellers on explore purchases. On average 60 % of a member's explore purchases are still outside the 300 candidates. Raising the cap to 500 recovers coverage (0.535) but not NDCG@20 on valid: the extra candidates are explore models the rankers cannot yet place in the top 20.

Tuning budget. The selected fits use 10 trials; a 30-trial re-tuning of every ranker changed valid NDCG@20 by at most 0.0004 and was not adopted. The final fits were repeated with three seeds: the seed spread ($\leq$ 0.0003 NDCG@20) is smaller than the P3 one-head gain over P1-tuned. The searches themselves ran with one sampler seed, and the candidate sources and NBR baselines kept the 10-trial budget; SASRec NBR has no NDCG@20 budget of its own (it keeps the R10 parameters, also when retrained on all targets). The margins between P1-tuned, P2 and P3 are below 0.001; P2's place depends on the budget (30 trials bring it level with P3 one head on test, not on valid), and only the gain of P3 one head over P1-tuned holds at every seed and at both budgets.

Offline only. GMV figures are a proxy; recommendations change behaviour, which an offline replay cannot measure.

Scope. SG labels are REVIEW; SG results are reported but not used for conclusions. gSASRec, distillation of P3 into a GBDT, conformal intervals, drift monitoring, and the what-if and parity check of the demonstrator were dropped from the plan.

Under-trained neural sources. The two-tower and SASRec sources gain 8 % explore coverage from 16 epochs instead of 8 (TH, valid); the comparison of neural and cheap sources is therefore at a fixed and binding epoch budget, and a rebuild with early stopping on the valid coverage is the first item of future work.

Future work. A full-catalogue universe with monthly refits addresses the coverage ceiling of the 5 000-model universe (84.5 % of the test positives); content embeddings of the catalogue text and the web click-stream as a second behaviour sequence, possibly with a generative sequential model such as HSTU [28], target the explore purchases that no co-purchase signal reaches.

X. CONCLUSION

Eleven operational exports became a point-in-time feature store on one machine: 66.1 M sale lines kept with every refused row traceable, 182 tests and zero leakage violations (RQ1). Measured over every purchase, the rule-based recommender built on it was half as good as the usual protocol says, and most of its misses were purchases it never saw. Treating retrieval as a model, with eight explore-only sources refitted quarterly and mixed under a 300-candidate cap, raised the share of purchases within reach from 0.370 to 0.450 (RQ2); with tuned rankers that see the source ranks, NDCG@20 rose from 0.095 to 0.1011–0.1020 (+7–8 %), more than 25 % ahead of the best next-basket baseline (0.0788). Features gave more than tuning, and tuning more than the model family: a deep two-head ranker with a learned gate does not beat a tuned GBDT, a single-head version does by 1 % (RQ3). The learned gate is essential; the explicit split is not. The best model is 1.8 times the best-sellers and 1.5 times "buy the same again" in one list of ten, and reaches about two thirds of the sales value; on purchase history alone the pipeline keeps about four fifths of its gain over the frequency rules, and the other data dimensions add the rest (RQ4). Naive stacking shows the inflation H1 predicted, small because the signals add little over a good feature store.

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